Gene Mining for Pesticide Targets in *Metisa plana*: A Computational Prioritisation Pipeline

Track 2 — Bio-Analyst (Undergraduate)

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1.0 Introduction

Metisa plana is a major oil palm leaf-defoliating pest in Malaysia, creating a need for targeted pest-management strategies. Its predicted proteome provides a useful resource for identifying candidate proteins, while recent *M. plana* research demonstrates the value of combining functional annotation, protein-domain analysis and transcriptomic evidence. However, sequence similarity alone cannot distinguish biologically relevant targets among homologous proteins. Therefore, this study aimed to identify and prioritise 10 candidate molecular targets from the *M. plana* predicted proteome by integrating sequence quality, homology, functional annotation, contamination screening, expression, phylogenetic support and target relevance.

2.0 Data and Strategy

The dataset contained 26,490 predicted M. plana proteins. The protein sequences were examined for evidence of homology, functional domains and evolutionary relationships, while gene identifiers were used to associate candidate proteins with developmental-stage expression data. Candidated classified into eight target classes: Chitin (100), Growth/Development (100), Digestion (95), Signaling (85), Detoxification (80), Transport (70), Transcription Regulation (65), and Other (30). RNA-seq expression data were available for pupal (SRR8905601), adult (SRR8905603), larval (SRR8905604 and SRR8905605), and egg (SRR8905607) stages.

The strategy prioritised candidates supported by multiple evidence types. Functional annotation assessed protein roles, taxonomic and evolutionary evidence evaluated biological context and relationships, while developmental-stage expression provided additional support for identifying proteins relevant to *M. plana* growth and survival. The RNA-seq data were particularly used to assess larval-stage expression, as the larval stage represents the primary feeding and damage-causing stage of *M. plana*. These evidence types were integrated to progressively prioritise proteins with plausible functional importance, target relevance and larval-stage relevance, ultimately producing a ranked list of the top 10 candidate targets

3.0 Methods

Proteins were screened for length, ambiguity and stop characters; terminal * were removed, internal stops retained. BLASTP hits were filtered at (≥30%) identity, (≥70%) query coverage and E-value ≤0.05 retaining top 10 hits. Taxonomic consistency was assessed, with arthropod matches preferred and bacterial/fungal matches flagged potential contamination before InterPro analysis as shown in Figure 3.0. For candidate mining, BLASTP hits were filtered using ≥30% sequence identity, ≥70% query coverage and E-value $\leq 1 \times 10^{-10}$. Strong hits were defined as ≥40% identity, ≥80% coverage and E-value $\leq 1 \times 10^{-20}$.

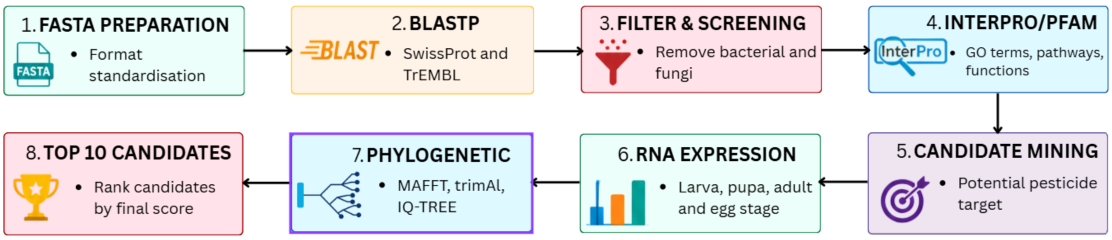

Figure 3.0 Computational workflow integrating BLAST homology, functional annotation, RNA-seq expression, and phylogenetic support to prioritise candidate targets.

Candidates were scored using homology (30%), taxonomy (25%), annotation (15%), InterPro (15%) and target relevance (15%). Candidates classified were ranked by final score, and the 500 highest-scoring proteins were retained for downstream RNA-expression prioritisation. RNA-seq was quantified using Salmon v1.10.2. TPM and NumReads were used to prioritise developmentally relevant candidates, with the top 100 retained for final integration.

Candidate and reference proteins were aligned using MAFFT, trimmed with trimAl and analysed by IQ-TREE to assess evolutionary placement and bootstrap support; unsupported subfamily assignments remained unassigned. Candidates were ranked using RNA expression (30%), homology (20%), functional annotation (20%), phylogeny (15%) and target relevance (15%). The 10 highest-ranked distinct genes were selected from the 500 → 100 → 10 workflow.

4.0 Results

4.1 RNA-seq Expression

All 10 final candidates exhibited their highest expression in one of the two third-instar larval datasets, with larval TPM values ranging from 113.62 to 1727.30. Egg and pupal expression levels were considerably lower for most candidates, supporting a predominantly larval-associated expression pattern.

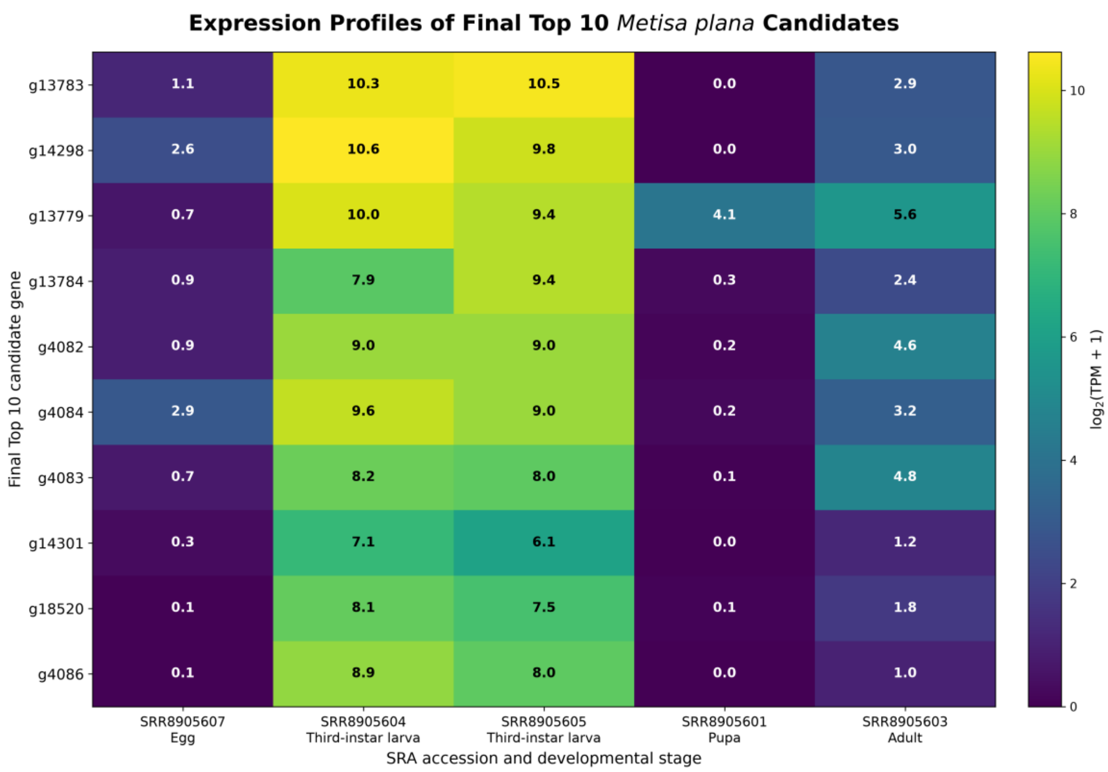

Figure 4.1 Expression profiles of the final Top 10 *Metisa plana* candidate genes across developmental stages. Values represent $\log_2(\text{TPM} + 1)$ from Salmon quantification of five public RNA-seq datasets. SRA accession numbers are provided to facilitate data tracking.

The expression heatmap showed that all 10 final candidates exhibited their highest expression in one of the third-instar larval datasets compared with the egg, pupal, and adult stages. Several candidates, including g13783, g14298, and g13779, showed particularly high expression in the third-instar larval datasets. Expression was generally higher in the SRR8905604 dataset for most candidates, while g13783 and g13784 showed higher expression in SRR8905605, and g4082 showed similar expression across both larval datasets. Overall, the strong larval expression pattern provided additional support for prioritising these candidates for further analysis.

4.2 Phylogenetic Support

Phylogenetic analysis was performed on all 10 final candidates. After trimming with trimAl, alignments retained 40–99% of their original length (mean 80.9%). IQ-TREE analysis produced mean UFBoot support values of 65.3–93.8% (mean 80.5%) and mean SH-aLRT values of 75.6–94.1% (mean 84.5%), indicating moderate to strong support for evolutionary placement. Phylogenetic support was considered alongside functional annotation when prioritising candidates.

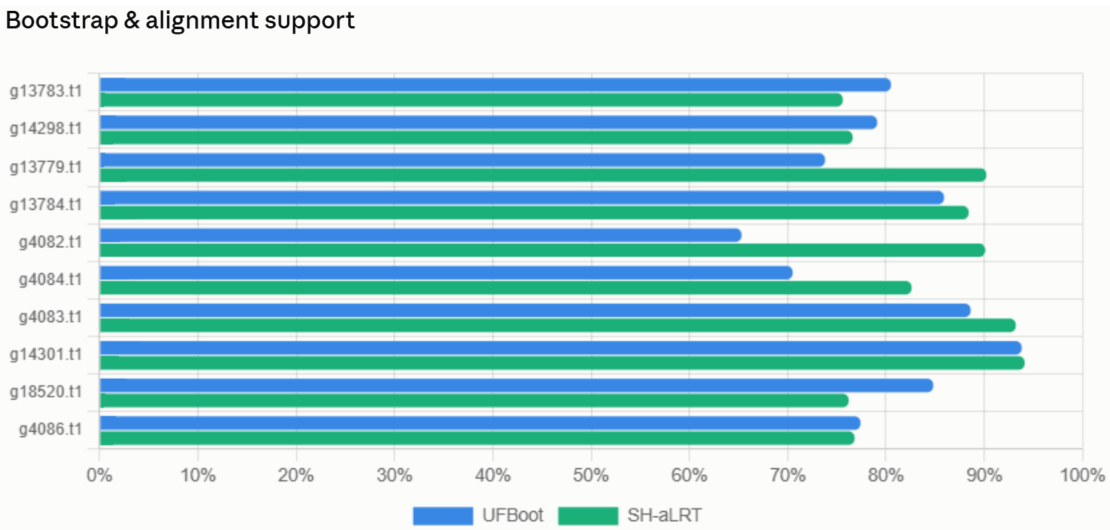

Figure 4.2: Phylogenetic support for the final 10 candidates, showing mean UFBoot (80.5%) and SH-aLRT (84.5%) values across candidate-specific trees.

Four trypsin candidates (g13783, g14298, g13779 and g13784) showed phylogenetic support consistent with their InterPro classification as Peptidase S1 family members. Lipase candidates also showed phylogenetic placement consistent with their functional annotations. Phylogenetic evidence contributed **15%** to the final integrated ranking, alongside RNA expression, homology, functional annotation and target relevance.

5.0 Discussion and Limitations

The workflow integrated four evidence types although interpretation is limited by the predicted proteome and computational reliance. SRR8905602 was excluded during QC due to unequal R1/R2 availability, and SRR8905603 was used as an alternative adult-stage dataset, potentially introducing sequencing differences. TPM-based expression estimates do not demonstrate differential expression, protein activity or gene essentiality. Similarly, computational RNAi or pesticide relevance does not establish experimental efficacy. Additionally, TPM values represent whole-larva or adult transcript abundance without tissue-specific resolution, so highly expressed genes in non-target tissues may have been prioritised despite low or absent expression in the midgut, the primary feeding organ for digestive proteases.

Phylogenetic support was generally moderate to strong, although g4082.t1 showed discordant support values (UFBoot 65%, SH-aLRT 90%), indicating some uncertainty in its evolutionary placement. The phylogenetic analysis was based on candidate-specific trees containing selected top BLAST homologs rather than comprehensive family-level phylogenies; therefore, broader subfamily relationships may not be fully resolved. Phylogenetic evidence was consequently used as one of five evidence types rather than as definitive functional confirmation. The 26,490 → 500 → 100 → 10 filtering pipeline may also exclude candidates early and cannot fully eliminate non-target sequences, so the final targets remain computational priorities requiring experimental validation.

6.0 Conclusion

This study established a reproducible computational workflow for prioritising potential pesticide targets in *Metisa plana* by integrating sequence quality, homology, contamination screening, functional annotation, expression, phylogeny and target relevance. The resulting Top 10 represent computationally supported candidates for further investigation. The workflow demonstrates how multiple evidence types can narrow a large, predicted proteome into a manageable set, while experimental validation remains necessary to confirm target efficacy.